

Teaching Exam

**Subject –Computer
Science**

Data handling Using Pandas



Lecture No.- 02

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Topics to be Covered

Topic

Data Frame



- ✓ Creation from various sources ✓
- ✓ Display and iteration ✓
- ✓ Row/column operations (add, select, delete, rename) ✓
- ✓ Head/tail functions ✓
- ✓ Indexing ✓
- ✓ CSV import/export ✓

Dictionary of Series

important pandas as pd.

Name = pd.Series(["Rahul", "Prakash", "Sanjiv"], index=["A", "B", "C"])

Age = pd.Series([24, 22, 23], index=["A", "B", "C"])

courses = pd.Series(["MCA", "BE", "BTech"], index=["A", "B", "C"])

Data = { 'Name': Name, 'Age': Age, 'courses': courses }

df = pd.DataFrame(Data)

print(df)

	Name	Age	courses
A	Rahul	24	MCA
B	Prakash	22	BE
C	Sanjiv	23	<u>NaN</u>
D	<u>NaN</u>	<u>NaN</u>	B.Tech

D
[1][2] →

`Data = [Name, Age, Courses]`

`df = pd.DataFrame(Data)`

`print(df)`

List of Servers

	A	B	C
0	Rahul	Prakash	Sanjay
1	24	22	23
2	MCA	BE	B.tech
:			

Dictonay of Dictionary

import pandas as pd

d1 = { "Name": 'Rahul', "Age": 24, "Course": 'MCA' }

d2 = { 'Prakash' 22, 'B.E' }

d3 = { 'Suryaz' 23, 'B.Tech' }

d4 = { "Operations": d1, "D-Marketing": d2, 'Sales': d3 }

df = pd.DataFrame(d4)

Print(df)

	Operations	Marketing	Sales
Name.	Rahul	Prakash	Suryaz
Age.	24	22	23
Course.	MCA	B.E	B.Tech

1. Creating DataFrames

From Dictionary of Series

python

```
import pandas as pd
```

```
data = {  
    'Math': pd.Series([90, 80, 70], index=['Alice', 'Bob', 'Charlie']),  
    'Science': pd.Series([85, 75, 65], index=['Alice', 'Bob', 'Charlie'])  
}
```

```
df = pd.DataFrame(data)
```

```
print(df)
```

	Math	Science
Alice	90	85
Bob	80	75
Charlie	70	65

b) From List of Dictionaries

```
data = [  
    {'Name': 'Alice', 'Age': 25}, ✓  
    {'Name': 'Bob', 'Age': 30}, ✓  
    {'Name': 'Charlie', 'Age': 35} ✓  
]
```

```
df = pd.DataFrame(data)  
print(df)
```

	Name	Age
0	Alice	25
1	Bob	30
2	Charlie	35

import pandas as pd.
import numpy as np

a = np.array([5, 8, 2, 10])

df = pd.DataFrame(a)

print(df)

	0
0	5
1	8
2	2
3	10

sa = pd.Series(a)

print(sa)

0	5
1	8
2	2
3	10

c) From Text/CSV File

Saving data to CSV first (for example)

df.to_csv("people.csv", index=False)

Reading from CSV

df_csv = pd.read_csv("people.csv")

print(df_csv)

	Name	Age
0	Alice	25
1	Bob	30
2	Charlie	35

→ print(df["Name"][1])

→ print(df.Name[1])

Accessing

Loc

print(df.at[1, "Name"])

print(df.iat[1, 0])

2. Display and Iteration

a) Display First and Last Rows

```
print(df.head(2)) # First 2 rows  
print(df.tail(1)) # Last 1 row
```

Output:

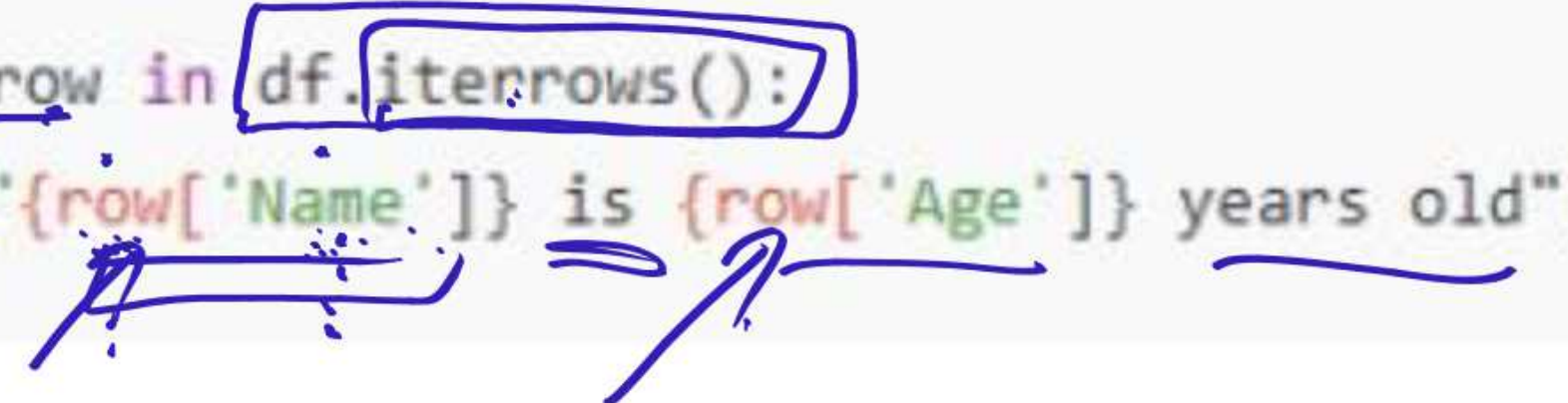
	Name	Age
0	Alice	25
1	Bob	30

	Name	Age
2	Charlie	35

b) Iterating Rows

python

```
for index, row in df.iterrows():  
    print(f"{row['Name']} is {row['Age']} years old")
```



Output:

pgsql

```
Alice is 25 years old  
Bob is 30 years old  
Charlie is 35 years old
```

3. Operations on Rows and Columns

python

```
df['Salary'] = [50000, 60000, 70000]  
print(df)
```

Output:

markdown

	<u>Name</u>	<u>Age</u>	<u>Salary</u>
0	Alice	25	50000
1	Bob	30	60000
2	Charlie	35	70000

b) Select Column(s)

```
print(df['Name'])           # Single column  
print(df[['Name', 'Age']]) # Multiple columns
```

Output:

pgsql

```
0    Alice  
1     Bob  
2  Charlie  
Name: Name, dtype: object
```

```
   Name  Age  
0  Alice  25  
1   Bob  30  
2 Charlie 35
```


c) Delete Column

```
df.drop('Salary', axis=1, inplace=True)  
print(df)
```

Output:

markdown

	Name	Age
0	Alice	25
1	Bob	30
2	Charlie	35

d) Rename Columns

```
df.rename(columns={ 'Name': 'FullName' }, inplace=True)  
print(df)
```

Output:

nginx

	FullName	Age
0	Alice	25
1	Bob	30
2	Charlie	35

loc, iloc

4. Indexing

a) Using Labels with `.loc[]`

```
print(df.loc[1])
```

Output:

```
vbnet
```

```
FullName      Bob
```

```
Age           30
```

```
Name: 1, dtype: object
```


b) Boolean Indexing

```
print(df[df['Age'] > 28])
```

Output:

nginx

	FullName	Age
1	Bob	30
2	Charlie	35

◆ 5. Importing/Exporting Data with CSV

✓ a) Export to CSV

python

```
df.to_csv("exported_file.csv", index=False)
```

→ This will create a file exported_file.csv in the current directory.

✅ b) Import from CSV

python

```
df2 = pd.read_csv("exported_file.csv")  
print(df2)
```

Output:

nginx

	FullName	Age
0	Alice	25
1	Bob	30
2	Charlie	35



2 mins Summary



TEACHING
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Topic

Dataframes

creation

various operations.

THANK YOU